REAL ANALYSIS MA337 FINAL EXAM. JANUARY 8, 2026.

1. (5 points) Let $C \subset [0,1]$ be the standard Cantor set, and $f(x)$ a function defined as $f(x) = \sin x$ for $x \notin C$, and $f(x) = 0$ otherwise. Prove that f is Riemann integrable on $[0,1]$, and determine the value of the integral.

Solution. Let $U := [0,1] \setminus C$. U is open, and f coincides on U with $g(x) = \sin x$, so f is continuous at every point in U. Hence f is bounded and almost everywhere continuous (since $\mu(C) = 0$), so by Lebegues criterion f is Riemann integrable.

The Riemann integral, if exists, coincides with Lebegues integral, but Lebegues integral is not sensitive to modifications on measure zero set, thus the desired Riemann integral equals

$$\int_0^1 \sin x dx = 1 - \cos 1.$$

2. (5 points) Is the function $f(x) := x \ln x$ (with $f(0) = 0$) absolutely continuous on $[0,1]$?

Solution. From Calculus we know that

$$x \ln x = \int_0^x (1 + \ln t) dt$$

(recall that the improper integral of $\ln t$ is finite due to the bound $|\ln t| < t^{-1/2}$ on $(0,1]$), and since the function has a definite sign in a neighborhood of 0, its Lebegues integral is finite too and coincides on each $[0,x]$ with the improper Riemann integral. We conclude that $f(x)$ is absolutely continuous, as the indefinite Lebegues integral of an L_1 function.

3. (5 points) Let $f(x)$ be a Lebegues measurable function on $[0,1]$, and

$$G = \{y = f(x)\} \subset \mathbb{R}^2$$

its graph. Prove that $\mu_2(G) = 0$, where μ_2 is the Lebegues measure in $\mathbb{R}^2$.

Solution. Note that $g(x,y) = f(x) - y$ is Lebegues measurable, as the difference of two meausurable, so G is measurable too (since it's the preimage of a point). This allows us to apply the "baby Fubini" Theorem, and we calculate

$$\mu_2(G) = \int_{[0,1]} 0 d\mu_x = 0$$

(each section S_x here consists of a single point, hence its 1-dimensional Lenegues measure is zero).

4. (7 points) Let $(\Omega, \mathcal{A}, \mu)$ be a space with a finite or σ-finite measure, and ν an absolutely continuous charge on $\mathcal{A}$ w.r.t. μ. Prove that the Radon-Nikodym derivative $d\nu/d\mu$ of ν w.r.t. μ is essentially unique, i.e. any two such derivatives are equal a.e.

Solution. Consider two Radon-Nikodym derivatives f, g of ν. Then for any $A \subset \mathcal{A}$ we have:

$$\int_A f d\mu = \int_A g d\mu.$$

We claim that the latter property implies $f = g$ a.e. Indeed, Ω is the union of the three sets A, B, C, where $A = \{f > g\}$, $B = \{f < g\}$, $C = \{f = g\}$, all measurable as f, g are measurable. Next, $A = \cup A_k$, $A_k = \{f - g > 1/k\}$. By Chebyshev inequality, $\mu(A_k) \leq k \int_{A_k} (f - g) d\mu = 0$, so $\mu(A_k) = 0$. Hence $\mu(A) = 0$. Similarly we prove that $\mu(B) = 0$, so Ω and C differ by a measure zero set, as desired.

5. (7 points) Consrtuct a partition

$$[0, 1] = \bigsqcup_n A_n,$$

where all A_n are measurable, and a measurable function f on $[0, 1]$ such that f is integrable on each A_n, the series

$$\sum_n \left| \int_{A_n} f d\mu \right|$$

is convergent, but $f \notin L_1([0, 1])$.

Solution. One can simply take $B_n = (1/(n + 1), 1/n]$ (and $B_0 = \{0\}$), and $f|_{B_n} := n(n + 1)(-1)^n$. We have $[0, 1] = \bigsqcup_n B_n$, $\int_{B_n} f d\mu = (-1)^n$. Then set

$$A_0 := B_0,\ A_1 := B_1 \cup B_2,\ A_2 = B_3 \cup B_4, ...$$

One computes, by definition of integral, that $\int_{A_n} f d\mu = 0$, which guarantees the convergence of the series, but $\int_{[0,1]} |f| d\mu = \sum \int_{B_n} |f| d\mu = +\infty$, so that f is not Lebegues integrable on $[0, 1]$.

6. (7 points) Prove the identity

$$\int_\Omega \left(\sum_{n=1}^{\infty} f_n(x) \right) d\mu = \sum_{n=1}^{\infty} \int_\Omega f_n(x) d\mu$$

(where f_n are nonnegative and measurable on a space with a complete measure $(\Omega, \mathcal{A}, \mu)$) without assuming the finiteness of neither the functions, nor the sum of functions, nor the integral, nor the sum of integrals.

This is the most general version of the Levi Theorem; your proof should be based on the standard (finite) version of the Theorem.

Solution. The problem is not difficult, but one has to accurately go case-by-case. Let's denote $f(x) := \sum_n f_n(x)$.

First, if some f_n is not a.e. finite, then, in view of $0 \leq f_n \leq f$, the same holds for f and hence both integrals of f_n and f are infinite and the identity holds trivially. So, we may assume from now that all f_n are a.e. finite.

Second, if the intergal of some f_n is infinite, by the same inequality the integral of f is infinite, and the identity holds trivially. So, we may assume now all the integrals of f_n are finite as well.

Thirdly, in view of $f_1 + ... + f_n \leq f$ we have $\int_\Omega f_1 d\mu + ... + \int_\Omega f_n d\mu \leq \int_\Omega f d\mu$, and if the sum of integral of f_n is infinite now, then the integral of f is automatically infinite too and the indentity holds. If, on the other hand, the sume of integrals is finite, by Levi theorem f is finite a.e., and the integral of f is the sum of integrals, as desired.

7. (10 points)

Let f_n be a sequence of functions in $L_p(\mathbb{R})$, $p \geq 1$ such that $\sum_n ||f_n||_{L_p} < +\infty$. Prove that the series $\sum_n f_n(x)$ is convergent in L_p and is absolutely convergent almost everywhere on $\mathbb{R}$.

Hint. First solve for $p = 1$.

Solution. Consider the sequence $g_n := (|f_1| + ... + |f_n|)^p$. Then we get a sequence of measurable nondecreasing functions (as obtained from measurable by arythmetic operations and composing with continuous from outside), and by Minkowski inequality:

$$\left(\int_\Omega g_n d\mu\right)^{1/p} = ||(|f_1| + ... + |f_n|)||_{L_p} \leq \sum_1^n ||f_j||_{L_p} \leq C,$$

where C is a constant (as follows from the given series convergence). Now, by Levi Theorem, g_n converges a.e., and so does the series $\sum |f_n|$.

Finally, to show the L_p convergence of the series, we argue by Cauchy criterion (using the completeness of L_p spaces), and use again the Minkowski inequality to estimate

$$||f_m + f_{m+1} + ... + f_n|| \leq \sum_m^n ||f_j||,$$

and now the convergence of the original series implies the fundamentality of the desired series in L_p and hence its convergence.

8. (10 points) Recall that every measurable $A \subset \mathbb{R}$ can be presented, for any $\epsilon > 0$, as $G \setminus E$ or $F \sqcup E'$, where G is open, F is closed, and $0 \leq \mu(E) \leq \epsilon$, $0 \leq \mu(E') \leq \epsilon$.

Is this assertion true for $\epsilon = 0$ and $\mu(A) > 0$?

Solution. We shall show that the assertion fails for $\epsilon = 0$. For doing so, we choose A to be a Cantor type set of measure $a \in (0, 1)$ (i.e. a closed nowhere dense subset of $[0, 1]$ with open dense complement U). Assume $A = G \setminus E$ where $\mu(E) = 0$ and G is open. E is nonempty, since A has no interior points, hence $G \cap U$ is a nonempty open set, so $\mu(G \cap U) > 0$ which implies $\mu(G) > \mu(A)$ and contradicts $\mu(G \setminus A) = 0$.

For the set U, the decomposition $U = F \sqcup E'$ with F closed and E' measure zero is impossible, because it implies $A = ([0, 1] \setminus F) \setminus E'$, which is impossible by the above.

9. (10 points) Construct a sequence of nonnegative functions $f_n \in L_1(\mathbb{R})$, uniformly converging on $\mathbb{R}$ to 0, but having integrals converging to the infinity.

Hint. First solve without requiring $f_n \in L_1(\mathbb{R})$.

Solution. A possible solution is considering $f_n(x) = 0$ for $|x| < n$ or $|x| > n^2$, and $f(x) = 1/|x|$ otherwise. Then we have $|f_n(x)| \leq 1/n$, so f_n converges to 0 uniformly on $\mathbb{R}$, but $\int_{\mathbb{R}} f_n(x)d\mu = 2\int_n^{n^2} dx/x = 2\ln n \to \infty$.

10. (10 points) Does there exist a sequence $G_n \subset \mathbb{R}^k$ of nonempty open sets with $\cup_n \partial G_n = \mathbb{R}^k$?

Solution. No, it doesn't exist, let's prove this. In view of the Baire category theorem, it's sufficient to show that each ∂G_n is nowhere dense in $\mathbb{R}^k$. Indeed, ∂G_n is closed, so if it's dense in some ball $B_r(a)$, then it contains $B_r(a)$. Since ∂G_n consists of boundary points of G_n, the ball $B_r(a)$ contains point in G_n, so there exist points a lying both in G_n and its boundary, which contradics the openness of G_n.

11. (10 points) Show the validity of Fatou theorem over spaces with a complete finite measure, if the almost everywhere convergence within it is replaced by convergence in measure.

Solution. We are given nonnegative measurable $f_n \in L_1(\Omega)$ converging in measure to a measurable f. We have to prove that $\int_\Omega f d\mu \leq \underline{\lim}_{n\to\infty} \int_\Omega f_n d\mu$.

Indeed, assume, by contradiction, that $\int_\Omega f d\mu > \underline{\lim}_{n\to\infty} \int_\Omega f_n d\mu$. By the defitinition of lower limit, choose f_{n_k} - a subsequence with $\lim_{k\to\infty} \int_\Omega f_{n_k} d\mu = \underline{\lim}_{n\to\infty} \int_\Omega f_n d\mu$. Then, using Riesz Theorem, we find a further subsequence $f_{n_{k_l}}$ converging a.e. to f. Applying to the latter subsequence the standard Fatou Theorem, we get

$$\int_\Omega f d\mu \leq \underline{\lim}_{l\to\infty} \int_\Omega f_{n_{k_l}} d\mu = \lim_{l\to\infty} \int_\Omega f_{n_{k_l}} d\mu = \lim_{k\to\infty} \int_\Omega f_{n_k} d\mu = \underline{\lim}_{n\to\infty} \int_\Omega f_n d\mu,$$

which gives us a contradiction.

12. (10 points) Let $f \in L_p(\Omega)$ $(p \geq 1)$ for a space $(\Omega, \mathcal{A}, \mu)$ with a finite complete measure. Prove that

$$\lim_{k\to\infty} k^p \cdot \mu\{x : |f(x)| \geq k\} = 0.$$

Solution. Let $A_k = \{x : |f(x)| \geq k\}$. Then $\cap A_k = \emptyset$, and by the continuity proprty of integral (a corollary of the σ-additivity) applicable in view of $f \in L_p(\Omega)$, $\lim_{k\to\infty} \int_{A_k} |f|^p d\mu = 0$. But $\int_{A_k} |f|^p d\mu \geq k^p \mu(A_k)$. Combining the two outcomes, we get the desired assertion.

13. (10 points) Prove that any function $f \in L_p(\mathbb{R})$ $(p \geq 1)$ can by $L_p(\mathbb{R})$-approximated by finite-valued simple functions from $L_p(\mathbb{R})$.

Solution. First, decompose $f = f^+ - f^-$ and conclude that it's emogh to approximate each $f^\pm$ and take the difference of the approximating functions. So, it's enough to approximate a nonnegative function f.

Second, putting $f_n := f \cdot \chi_{[-n,n]}$, we have $f_n \geq 0$ converging a.e. monotoneously to f. By Levi Theorem applied to $-(f - f_n)^p$, we conclude that f_n convergence in $L_p(\mathbb{R})$ to f. Note that f_n vanishes outside $[-n, n]$. Hence it's now sufficient to approximate a nonnegative function f defined on $[-A, A]$.

Thirdly, since $[-A, A]$ has the finite measure, applying the definition of integral, we find a sequence $g_n \in L_p([-A, A])$ of (infinite-valued) simple functions converging to f uniformly on $[-A, A]$. Since $[-A, A]$ has the finite measure, the uniform convergence implies the convergence of g_n to f in $L_p([-A, A])$. Wo everything is reduced now to a nonnegative simple function from $L_p([-A, A])$.

Finally, since $f = \sum_{n=1}^{\infty} c_n \chi_{E_n}$, we define $h_m := \sum_{n=1}^{m} c_n \chi_{E_n}$, and $\int_{[-A,A]} (f - h_m)^p = \sum_{n=m+1}^{\infty} c_n^p \mu(E_n) \to 0$ as $m \to \infty$ since $f \in L_p([-A, A])$.

This proves the claim.

14. (10 points) Does every continuous function on $[0, 1]$ have bounded variation?

Solution. No. Let's consider e.g. the function $f(x) := x \sin(1/x)$ and $f(0) = 0$. Since $|f(x)| \leq |x|$, the limit of f at 0 is 0 and $f \in C[0, 1]$. Taking now the partition $x_0 = 1, x_1 = 2/\pi, x_2 = 2/(3\pi), x_3 = 2/(5\pi), ...,$ up to step $n - 1$ and finally $x_n = 0$, we obtain a variational sum of the kind $\sum_{k=1}^{n} O(1/k)$. The respective series is divergent to $+\infty$, thus variational sums are unbounded from above and f is not of bounded variation.

15. (15 points) Let $f \in L_1([a, b])$, and φ be a strictly increasing absolutely continuous on $[\alpha, \beta]$ function s.t. $\varphi([\alpha, \beta]) = [a, b]$. Prove the change of variable principle

$$\int_{[a,b]} f(x) d\mu = \int_{[\alpha,\beta]} f(\varphi(t))\varphi'(t) d\mu,$$

where μ is the standard Lebegues measure, and φ' stands for the a.e. derivative.

Solution. Let $F(x) := \int_{[0,x]} f(y) d\mu$. Then $F(x)$ is absolutely continuous, and $F'(x) = f(x)$ a.e. Let $G(t) := F(\varphi(t))$. It's easy to argue that G is also absolutely continuous on $[\alpha, \beta]$: indeed, for any $\epsilon > 0$ we choose $\delta = \delta(\epsilon)$ according to the absolute continuity property of F and $\tau = \tau(\delta)$ according to the absolute continuity property of φ. Then any system of disjoint intervals S with total length $< \tau$ is mapped by φ into such with total length $< \delta$, hence by F into such with total length $< \epsilon$.

Next, applying pointwise (!) the chain rule, we get $G'(t) = F'(\varphi(t)) \cdot \varphi'(t) = f(\varphi(t)) \cdot \varphi'(t)$ a.e.. Now by the Newton-Leibnitz formula for absolutely continuous functios, we see that both the LHS and the RHS equal $F(b) - F(a) = G(\beta) - G(\alpha)$.

Remark. A solution via approximation by simple functions requires full details, including the proof of existence of the RHS. This can be argued by employing the Levi Theorem.

16. (15 points) Let $f \in L_1(\mathbb{R})$ w.r.t. the standard Lebegues measure μ. Prove that $\lim_{a\to 0} \int_{\mathbb{R}} |f(x + a) - f(x)| d\mu = 0$.

Hint. Use approximation by appropriate functions.

Solution. We note the following: if the desired property holds for a sequence f_n converging to f in $L_1(\mathbb{R})$, then it holds for f too. This follows from the inequality

$$\int_{\mathbb{R}} |f(x + a) - f(x)| d\mu \leq \int_{\mathbb{R}} |f(x + a) - f_n(x + a)| d\mu + \int_{\mathbb{R}} |f_n(x + a) - f_n(x)| d\mu +$$

$$+\int_{\mathbb{R}} |f_n(x) - f(x)| d\mu = 2||f - f_n||_{L_1} + \int_{\mathbb{R}} |f_n(x+a) - f_n(x)| d\mu.$$

We next apply e.g. the outcome of Problem 13 (or argue identically to Problem 13) to find bounded L_1-approximating functions vanishing outside $[-A, A]$ (they are actually simple, but we don't need this property). Finally, for a bounded f vanishing outside $[-A, A]$, we find for each $\epsilon > 0$ by Lusin Theorem a continuous function g on $\mathbb{R}$ with $\mu\{f \neq g\} < \epsilon$ and also $sup|g| \leq sup|f|$ (as follows from the construction of the Lusin function). Then we compute $||f - g||_{L_1} < C\epsilon$, $C = 2sup|f|$. This means the original function can be L_1-approximated by continuous ones supported on $[-A, A]$. But those clearly have the desired property by their equicontinuity on $\mathbb{R}$. This gives the desired claim.